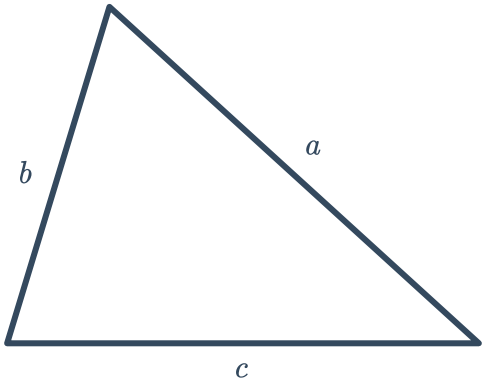
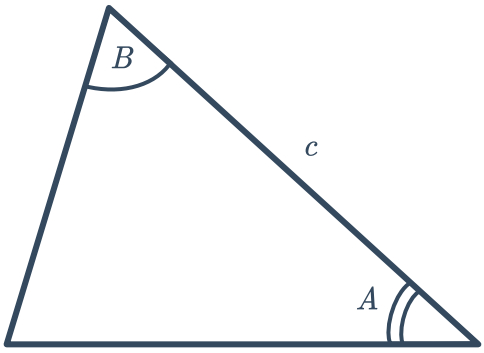
Sine rule

1 For each of the given triangles, determine if there is enough information to find all the remaining sides and angles in the triangle using only the sine rule:

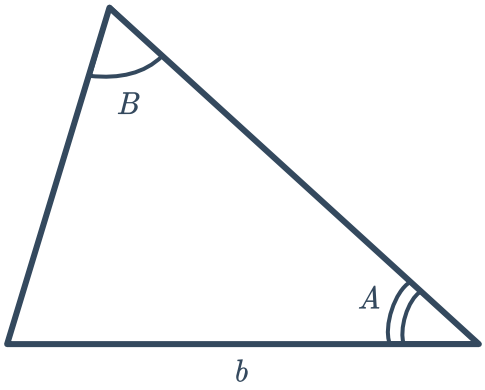
a Three sides are known:



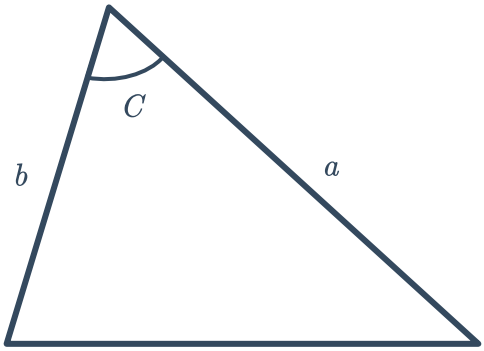
b Two of the angles and the side included between them are known:



c Two of the angles and a side not included between them are known:



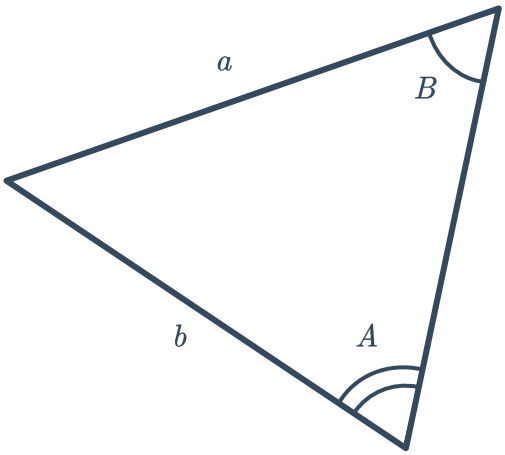
d Two of the sides and an angle included between them are known:



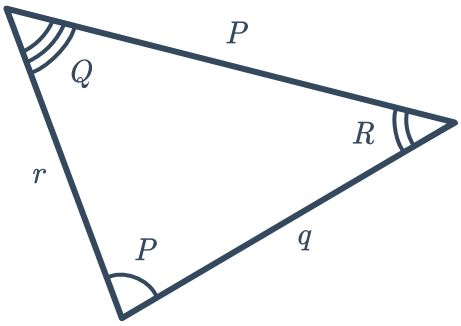
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2 For each of the following triangles, write an equation relating the sides and angles using the sine rule:

a



b

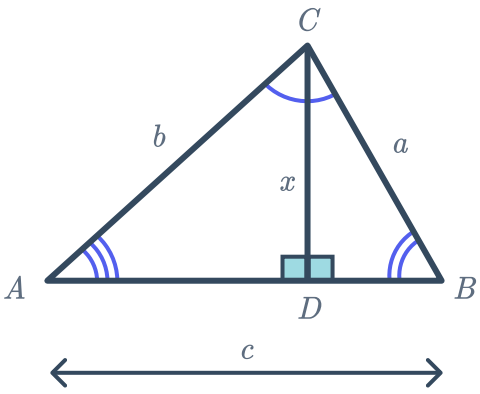


3 Consider the following diagram:

a Find an expression for $\sin A$ in $\triangle ACD$.

b Find an expression for $\sin B$ in $\triangle BDC$. Then make x the subject of the equation.

c Substitute your expression for x into the equation from part (a), and rearrange the equation to form the sine rule.

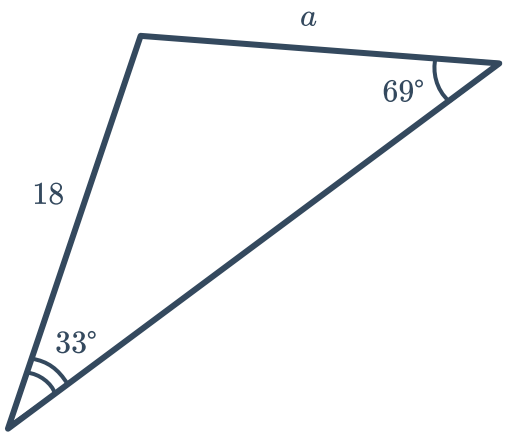


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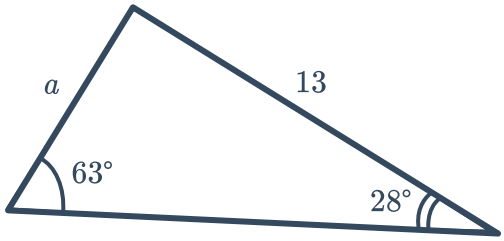
Unknown sides

4 For each of the following triangles, find the side length a using the sine rule. Round your answers to two decimal places.

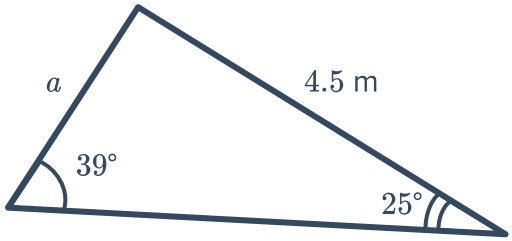
a



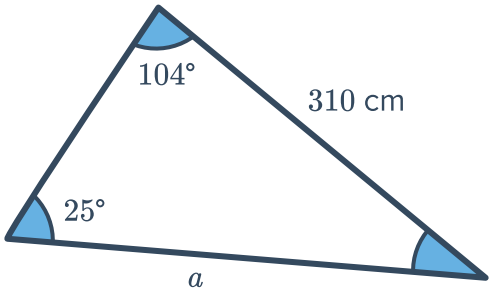
b



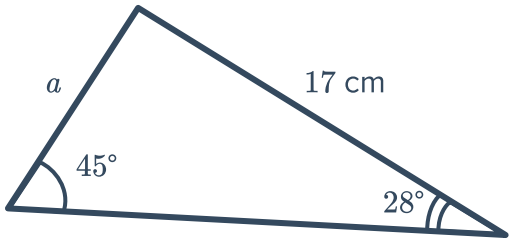
c



d



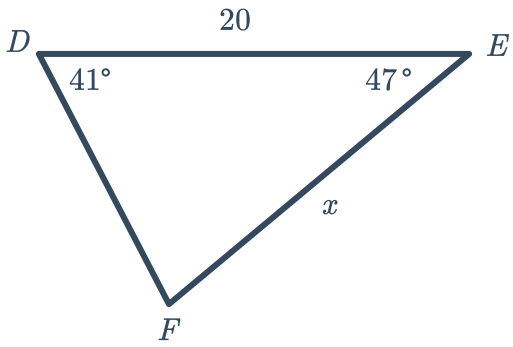
e



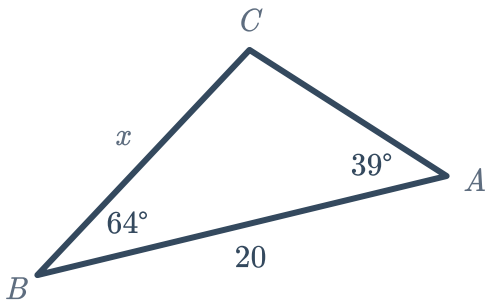
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5 For each of the following triangles, find the length of side x, correct to one decimal place:

a



b

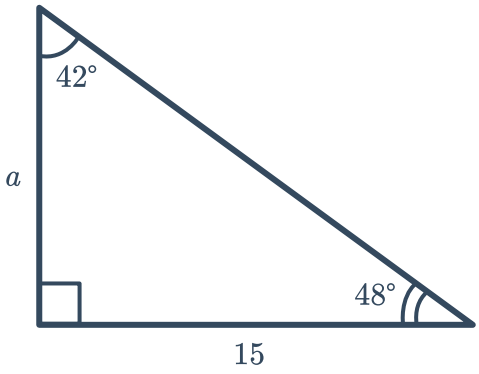


6 For each of the following triangles:

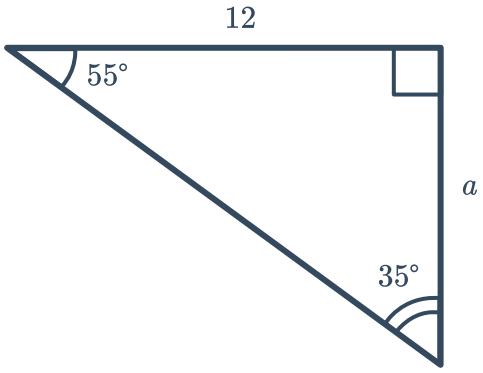
i Find the value of a using the sine rule. Round your answer to two decimal places.

ii Use another trigonometric ratio and the fact that the triangle is right-angled to calculate and confirm the value of a. Round your answer to two decimal places.

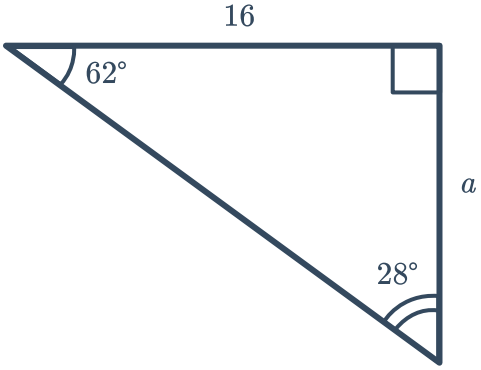
a



b



c

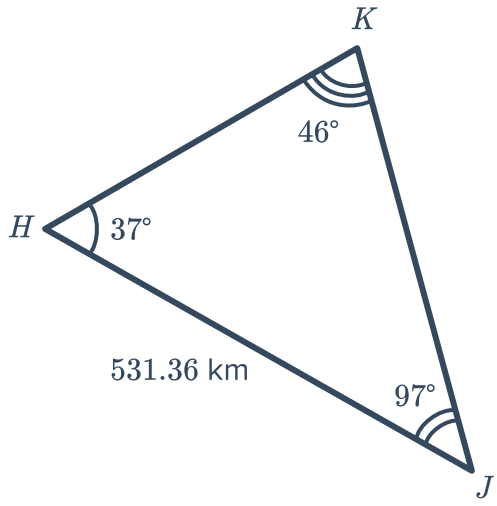


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7 Consider the following triangle:

a Find the length of side HK to two decimal places.

b Find the length of side KJ to two decimal places.

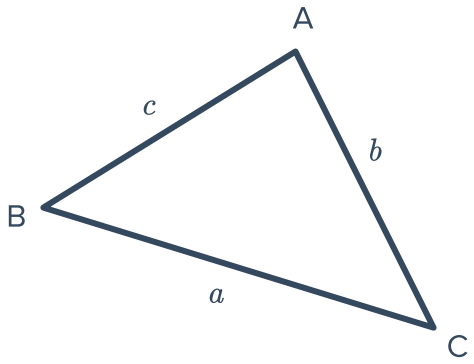


8 Consider the triangle with $\angle C = 72.53^\circ$ and $\angle B = 31.69^\circ$, and one side length $a = 5.816$ m.

a Find $\angle A$. Round your answer to two decimal places.

b Find the length of side b. Round your answer to three decimal places.

c Find the length of side c. Round your answer to three decimal places.

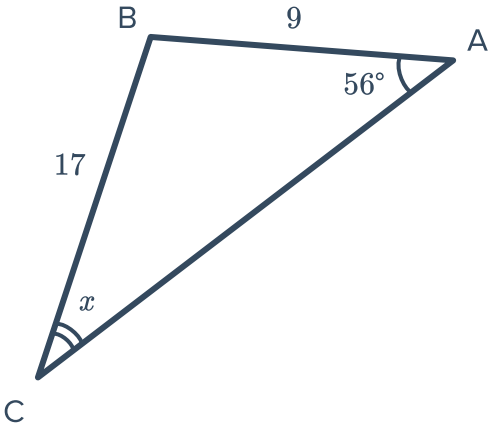


Unknown angles

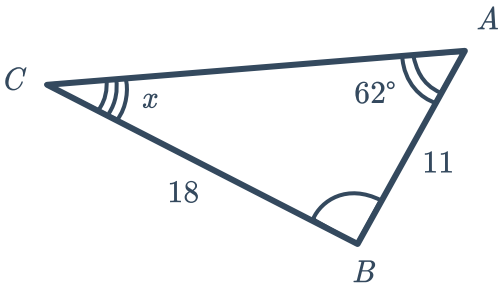
9 For each of the following diagrams, find the value of the angle x using the sine rule. Round your answers to one decimal place.

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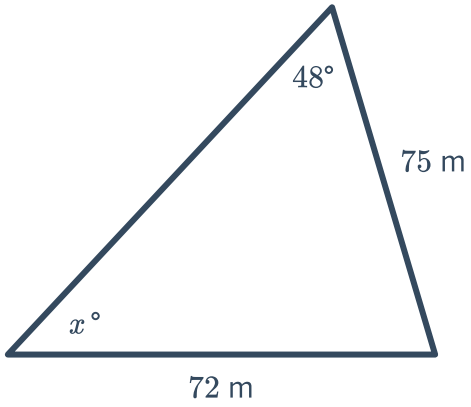
a



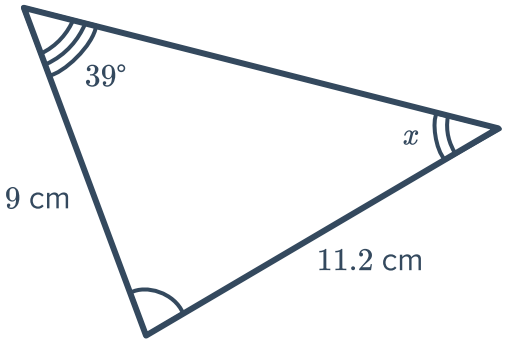
b



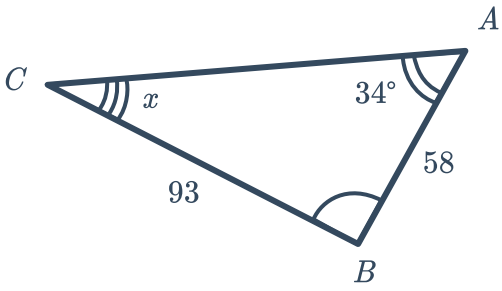
c



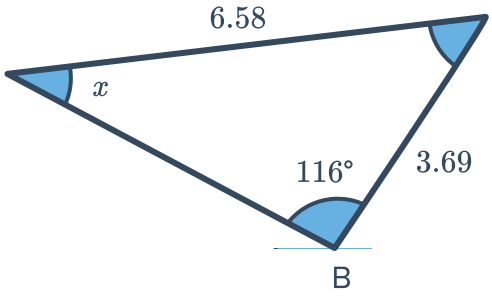
d



e

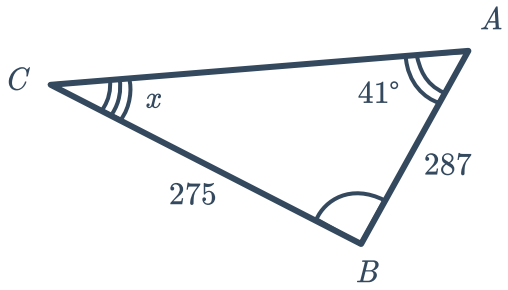


f

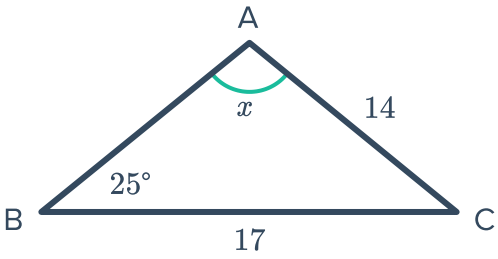


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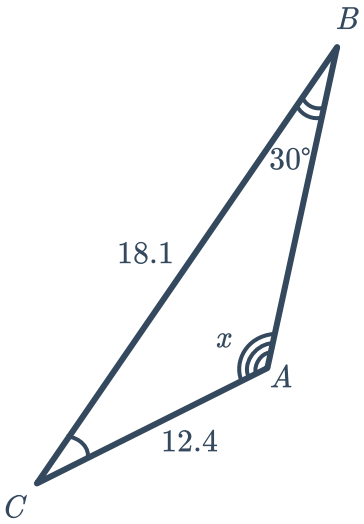
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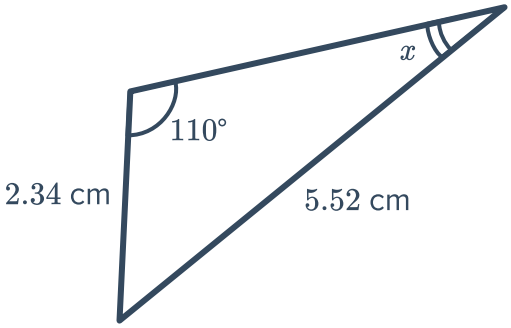
h



i



j



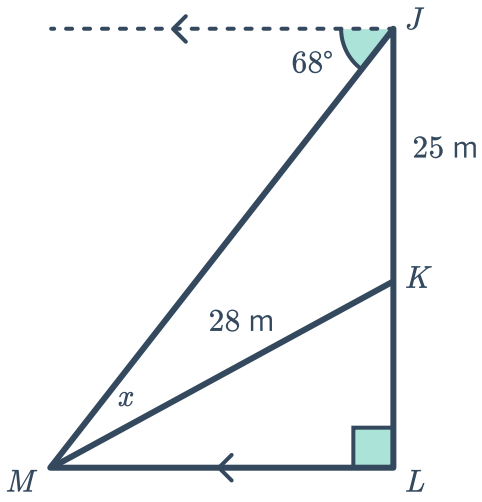
10 For each of the following acute angled triangles, calculate the size of $\angle B$ to the nearest degree:

a $\triangle ABC$ where $\angle A = 57^\circ$ side $a = 156$ cm and side $b = 179$ cm

b $\triangle ABC$ where $\angle A = 48^\circ$ side $a = 2.7$ cm and side $b = 1.9$ cm

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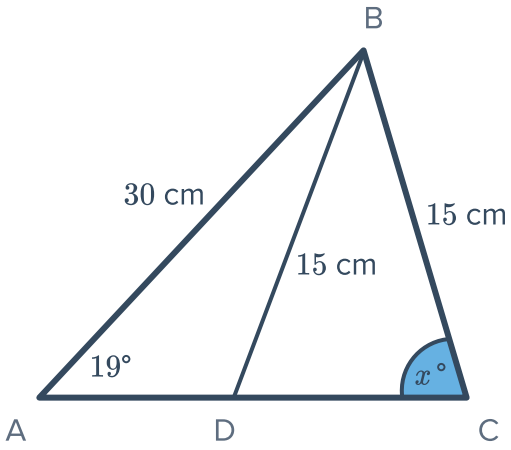
11 The angle of depression from J to M is 68°. The length of JK is 25 m and the length of MK is 28 m as shown:
Find the following, rounding your answers to two decimal places:



a Find x, the size of $\angle JMK$.

b Find the angle of elevation from M to K.

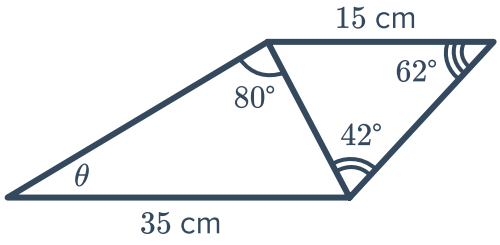
12 Consider $\triangle ABC$ below:



a Find x to the nearest degree.

b Find $\angle ADB$ to the nearest degree.

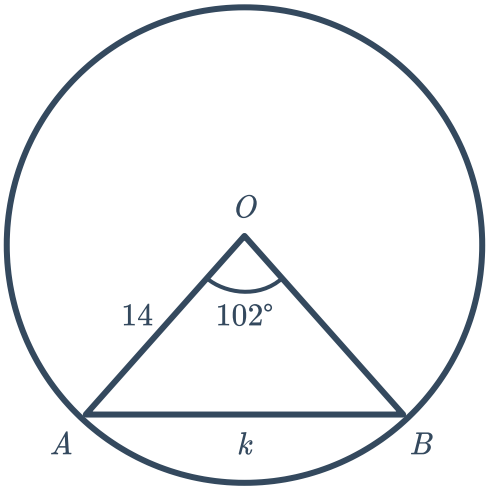
13 Consider the following diagram of a quadrilateral:
Find the value of θ, correct to two decimal places.



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Applications

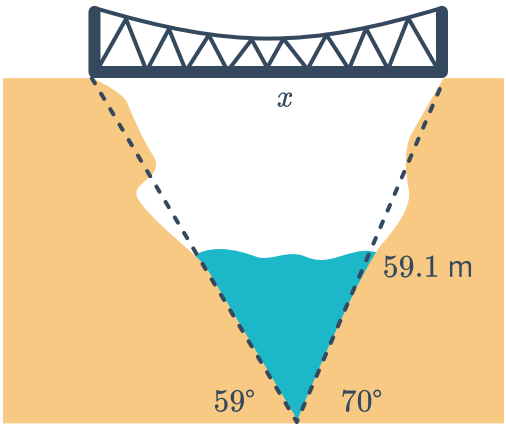
14 Consider the following diagram:



a Find the size of $\angle OBA$.

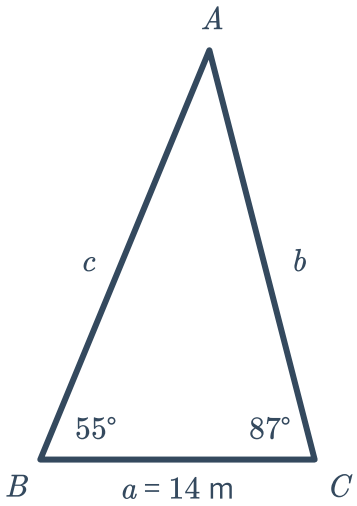
b Find the length of k, to two decimal places.

15 A bridge connects two towns on either side of a gorge, where one side of the gorge is inclined at 59° and the other side is inclined at 70°. The length of the steeper incline is 59.1 m.
Find x, the length of the bridge. Round your answer correct to one decimal place.



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16 During football training, the coach marks out the perimeter of a triangular course that players need to run around. The diagram shows some measurements taken of the course, where side length $a = 14$ m:



a Find the size of $\angle A$.

b Find the length of side c, correct to two decimal places.

c Find the length of side b, correct to two decimal places.

d Each player must sprint one lap and then jog one lap around the triangle. This process is to be repeated 3 times by each player.
If Tara can run 280 m/min, and can jog at half the speed she runs, calculate the time this exercise will take her, correct to one decimal place.

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